AlphaFree: Recommendation Free from Users, IDs, and GNNs (Online Appendix)

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1 Computational Complexity Analysis

Time complexity analysis. We theoretically analyze the time complexity of each phase of AlphaFree, where Algorithms 1, 2, and 3 show the preprocessing, training, and inference phases, respectively. Suppose $n = |\mathcal{I}|$ is the number of items, $m = \sum_{I \in \mathcal{H}_{\text{train}}} |I|$ is the total number of interactions, $h = |\mathcal{H}_{\text{train}}|$ is the number of interaction sets ($h \leq m$), K_c is the number of similar items ($K_c \leq n$), $|\mathcal{B}|$ is the size of mini-batch, d_{LR} is the dimension of LRs, and d denotes the embedding dimensions of the MLP layers.

Lemma 1 (Time Complexity for Preprocessing). *The time complexity of Algorithm 1 is $O(T_{\text{LM}}n + mn + (d_{\text{LR}} + \log K_c)n^2)$, where T_{LM} denotes the time at which the LM produces an embedding.*

Proof. The algorithm takes $O(T_{\text{LM}}n)$ time as it computes the LRs of all items (line 1). Constructing an inverted list for $H(i)$ takes $O(m)$ time, since $H(i)$ for each i is built by iterating over all interactions in $\mathcal{H}_{\text{train}}$ (line 2). Each intersection for $\texttt{sim}_B(\cdot)$ takes $O(|H(i)| + |H(j)|)$ time, assuming each $H(i)$ is sorted, and then it takes $\sum_{i \leq j}(|H(i)| + |H(j)|) = (n+1)\sum_i |H(i)| = O(nm)$ time (line 6). Computing all $\texttt{sim}_S(\cdot)$ takes $O(d_{\text{LR}}n^2)$. Then, it consumes $O(n^2 \log K_c + nK_c)$ time to find similar items $\mathcal{S}_i$ for all items (line 8), if we use min-heap for sorting. It takes $O(mK_c)$ and $O(nK_c d_{\text{LR}})$ for obtaining I^+ and $\mathbf{z}_i^+$, respectively (lines 9 - 11). Combining all the above terms proves the claim. ∎

Lemma 2 (Time Complexity for Training). *In Algorithm 2, each epoch takes $O((T_i dn + K_c m + dh)d_{\text{LR}} + (n_s + |\mathcal{B}|)hd)$ time, where $T_i = h/|\mathcal{B}|$ is the number of iterations over all batches, and n_s is the number of negatives.*

Proof. We first analyze the time for processing each mini-batch $\mathcal{B}$. It takes $O(d_{\text{LR}}dn)$ time to perform the MLP operations for all items (line 4). It takes $O(d_{\text{LR}}(|I|+d))$ time for each I and $O(d_{\text{LR}}(|I^+|+d))$ for each I^+ (lines 6 and 7), where $|I^+| = O(K_c|I|)$, thus the overall cost becomes $O(d_{\text{LR}}(K_c|I| + d))$. Sampling p and N_I takes $O(n_s)$ time in expectation when using rejection sampling (line 8). Then, it takes $O(|\mathcal{B}|n_s d)$ time for $\mathcal{L}_{\text{rec}}$, and $O(|\mathcal{B}|^2 d)$ time for $\mathcal{L}_{\text{align}}$ (lines 9 and 10). Considering all batches is represented as follows:

$$\begin{aligned}
&\sum_{\mathcal{B} \subseteq \mathcal{H}_{\text{train}}} \Big(d_{\text{LR}}dn + \sum_{I \in \mathcal{B}} d_{\text{LR}}(K_c|I| + d) + n_s + |\mathcal{B}|n_s d + |\mathcal{B}|^2 d \Big) \\
&\quad = T_i d_{\text{LR}}dn + d_{\text{LR}}K_c m + d_{\text{LR}}dh + T_i n_s + hn_s d + |\mathcal{B}|hd \\
&\quad = O((T_i dn + K_c m + dh)d_{\text{LR}} + (n_s + |\mathcal{B}|)hd,
\end{aligned}$$

where $n_s \leq n$. ∎

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Lemma 3 (Time Complexity for Inference). *The time complexity of Algorithm 3 is $O(d_{\text{LR}}dn + n \log K)$, where K is the parameter for top-K recommendation.*

Proof. The algorithm takes $O(d_{\text{LR}}(|I_q| + d))$ time to perform the averaging and MLP operations (line 1). For all items, it takes $O(d_{\text{LR}}dn)$ for computing their embeddings (line 2). Then, it takes $O(dn + n \log K)$ time to compute the scores of all items and select the top-K items using a min-heap (line 3). Combining all the above terms proves the claim. ∎

Space complexity analysis. We analyze the space complexity of each phase.

Lemma 4 (Space Complexity for Preprocessing). *The preprocessing phase of AlphaFree consumes $O(d_{\text{LR}}n + K_c(m+n))$ space.*

Proof. The LRs $\{\mathbf{z}_i\}$ and $\{\mathbf{z}_i^+\}$ for all items i require $O(d_{\text{LR}}n)$ space (lines 1 and 11). The $\{H(i)\}$ for all items uses $O(m)$ space (line 2). The set I^+ uses $O(K_c|I|)$ space, and thus for all $I \in \mathcal{H}_{\text{train}}$, it takes $O(K_c m)$ space (line 10). To find and store $\mathcal{S}_i$, it uses $O(K_c)$ space when a heap is used, and $O(K_c n)$ in total for all items (line 8). Note that for each i, it computes similarity scores of i for every j in an in-place manner, requiring only $O(n)$ space without storing results from previous i's (lines 5-7). Combining all the above terms proves the claim. ∎

Lemma 5 (Space Complexity for Training). *The training phase of AlphaFree uses $O(m + d(n + d_{\text{LR}} + n_s + |\mathcal{B}|))$ space.*

Proof. As an input, $\mathcal{H}_{\text{train}}$ uses $O(m)$ space. Each MLP layer takes $O(d_{\text{LR}}d)$ space (line 1). The hidden embeddings $\{\mathbf{e}_i\}$ and $\{\mathbf{e}_i^+\}$ use $O(dn)$ space (line 4). For line 9, it uses $O(dn_s)$ space for $\mathcal{E}_I$ and $\mathcal{E}_I^+$ used to compute the local losses for $\mathcal{L}_{\text{rec}}$ in an in-place manner. For line 10, it uses $O(|\mathcal{B}|d)$ space for $\mathcal{E}_{\mathcal{B}}$, $\mathcal{E}_{\mathcal{B}}^+$, $\mathcal{E}_{P_{\mathcal{B}}}$, and $\mathcal{E}_{P_{\mathcal{B}}}^+$ used to compute the local losses for $\mathcal{L}_{\text{align}}$ in an in-place manner. Combining all the above terms proves the claim. ∎

Lemma 6 (Space Complexity for Inference). *The inference phase of AlphaFree uses $O((d_{\text{LR}} + d)n + K)$ space.*

Proof. Computing all $\{\mathbf{e}_i\}$ as well as $\mathbf{e}_{I_q}$ takes $O((d_{\text{LR}} + d)n)$ space (line 2), while selecting the top-K items requires $O(K)$ space if a heap is used (line 3). ∎

2 Case Study for Over-Smoothing

Figure 1 shows a case study comparing AlphaRec and AlphaFree to investigate the over-smoothing issue on the Baby dataset. For the heavy user (User A) with many interactions ($|I_A| = 191$), AlphaRec assigns uniformly high scores with very low variance, suggesting that its item embeddings have become excessively similar (i.e.,

over-smoothed) due to GNNs. However, AlphaFree constructs user representations without GNNs, thereby preserving embedding diversity and avoiding over-smoothing for User A. For the light user (User B) with few interactions ($|I_B| = 17$), AlphaRec yields higher yet broadly distributed scores than AlphaFree, indicating that even under sparse interactions, it produces more similar but unstable embeddings, whereas AlphaFree shows less similarity with a more compact and stable distribution, suggesting robustness against such effects.

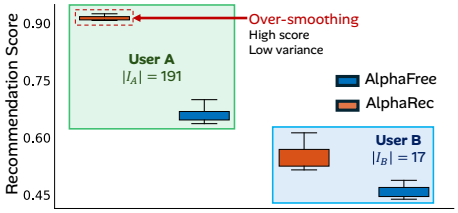

Figure 1: Case study of over-smoothing of AlphaRec and AlphaFree for heavy and light users on Baby.

3 Correction of Inference Time

After the camera-ready submission, we identified an issue in the implementation of the inference-time measurement reported in the published paper. We therefore correct the issue and report the newly measured inference time in this material. Specifically, during the inference phase, we measure the average time T_i required to generate recommendations for a single user, reported in microseconds. Table 1 shows the results in running time. From the inference-time perspective, AlphaFree achieves latency comparable to LightGCN. In contrast, AlphaRec consistently exhibits higher inference latency. It is worth noting that both AlphaRec and AlphaFree operate on the dimension d_{LR} of LRs exceeding 3,000, whereas LightGCN performs computations using 64-dimensional embeddings.

Table 1: Runtime comparison of AlphaFree with LR-based methods and LightGCN. T_p (sec), T_t (sec), and T_i (μs) indicate preprocessing, training (per epoch), and inference (per query user) times, respectively. Parentheses under T_p for LR-based methods indicate the ratio for generating LRs during preprocessing.

Methods	RLMRec			AlphaRec			AlphaFree			LightGCN		
Datasets	T_p	T_t	T_i	T_p	T_t	T_i	T_p	T_t	T_i	T_p	T_t	T_i
Video	o.o.t.	n/a		11(98%)	5	19	563(98%)	9	14	10	1	13
Baby	o.o.t.	n/a		797(98%)	12	23	802(98%)	13	18	39	1	18
Steam	o.o.t.	n/a		342(98%)	91	28	315(98%)	45	3	84	8	9
Beauty	o.o.t.	n/a		4614(99%)	n/a	(o.o.m.)	4612(99%)	157	107	92	36	97
Health	o.o.t.	n/a		4211(99%)	n/a	(o.o.m.)	4180(99%)	161	93	563	39	90